

ACTIVATION SPARSIFICATION IN TRANSFORMERS: PRESSURE, THRESHOLDING, AND SITE PLACEMENT

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ABSTRACT

Useful activation sparsity depends on how zeros are induced, which operations they affect, and whether exploiting them repays execution overhead. We study pressure, thresholding and site placement through 54 matched Pythia pretraining conditions on MiniPile: detailed comparisons at 14M and selected complete recipes at 70M and 410M. Model-wide activation sparsity, S_{model} , counts zero-operand scalar multiplications in a declared workload; architectural reach, R_{arch} , describes the portion structurally affected by the selected sites. Pressure’s marginal quality–sparsity effect changes with the threshold and target objective, from improving both observed axes to exchanging quality for additional zeros. Distribution measurements and operation counts also show that the seven-site recipe can produce greater model-wide sparsity despite fewer pooled FFN zeros, because its attention contribution outweighs reduced projection contributions. The high-threshold complete-recipe advantage persists across the three sizes, but lower-threshold orderings vary. On one RTX5090, a selected implementation achieves a $1.234\times$ geometric mean speedup over native execution across the 30 14M checkpoints for BF16, uncached 2,048-token inference. Sparse paths add $1.043\times$ over matched fusion, helping 14 checkpoints; attention skipping slows every checkpoint. These single-seed comparisons motivate evaluating an intervention’s paired quality cost, affected operations and execution granularity together.

1 INTRODUCTION

Activation sparsity can reduce inference cost when specialized kernels exploit zero activations (Liu et al., 2024; Cetin et al., 2026). Choosing how to create those zeros requires more than a local sparsity percentage: their sites determine the affected multiplication work, and skipping that work adds overhead. We ask how pressure, thresholding and placement change the quality cost of sparsification, and which opportunities translate into faster execution.

Existing recipes combine these choices differently: TEAL uses post-hoc thresholding with allocation across projections; ProSparse combines ReLU, progressive pressure and threshold shifting; Q-Sparse selects top- k projection inputs (Liu et al., 2024; Song et al., 2024; Wang et al., 2024). Their ablations motivate examining the additional value of pressure as the threshold and sites change. We distinguish *pressure*, which encourages small magnitudes, from *thresholding*, which applies specified zeroing maps, and *placement*, which selects where each acts.

We pretrain Pythia-family models from scratch for controlled comparisons across sizes (Biderman et al., 2023). At 14M, 30 conditions and 29 paired comparisons show that pressure’s marginal effect depends on the threshold. At the largest threshold, adding orthogonal pressure to seven-site thresholding raises model-wide sparsity from 15.39% to 27.48%, against 29.95% architectural reach, for $+0.1265$ loss relative to the unpressured recipe and $+0.6208$ relative to unmodified A0. This recipe has greater model-wide sparsity than its four-site counterpart despite fewer pooled FFN zeros; operation counts explain the reversal. Selected recipes extend to 70M and 410M, where the high-threshold seven-site advantage persists but lower-threshold orderings vary. Uniform post-hoc clipping provides a supporting comparison: small quality budgets can favor clipped endpoints.

A human-guided coding-agent case study then measures execution on one RTX5090, using BF16, batch-one, uncached 2,048-token inference. The selected implementation is $1.234\times$ faster than na-

PyTorch/SDPA geometrically across all 30 14M checkpoints. Sparse paths add $1.043\times$ over matched fusion with skipping disabled, helping 14 checkpoints; disabling attention skipping improves every checkpoint. These comparisons connect an intervention’s quality cost and affected operations to the instruction granularity and overhead that determine whether its zeros are useful.

2 RELATED WORK

2.1 SPARSIFICATION INTERVENTIONS

CATS thresholds small FFN activations in training-free and fine-tuning settings (Lee et al., 2024); TEAL extends magnitude thresholding to attention and FFN projections with block-wise greedy allocation (Liu et al., 2024). Our uniform clipping reference does not test TEAL’s optimized allocation. Training can also reshape the trade-off: ReLU Strikes Back restores sparse activations (Mirzadeh et al., 2023); ProSparse combines ReLU, progressive ℓ_1 pressure and threshold shifting, with schedule ablations showing different quality at comparable FFN sparsity (Song et al., 2024). Sparsing Law studies data, activation and architecture effects (Luo et al., 2025), and Cetin et al. (2026) induce sparse FFNs through pretraining regularization.

Beyond FFNs, Q-Sparse applies top- k selection to attention and FFN projection inputs and ablates its nonlinearity and straight-through estimator (Wang et al., 2024). Spark combines sparse FFNs with attention-position selection and studies its predictor (You et al., 2025). Our seven-site intervention instead also thresholds feature coordinates inside attention. We examine pressure’s conditional effect and the resulting operation-weighted counts, with and without pressure at 14M.

2.2 INFERENCE SPEEDUP

TEAL and Spark report autoregressive decoding gains (Liu et al., 2024; You et al., 2025). TwELL targets batched matrix multiplication with packed sparse activations and fused FFNs, evaluating full-sequence throughput (Cetin et al., 2026). Workload and structure matter: different token masks affect batching savings; Block Q-Sparse structures masks for batched execution (Wang et al., 2024). Our case study uses execution and profiling feedback for kernel development (Ouyang et al., 2025; Luo et al., 2026), then matched fusion and skipping controls to identify what produces full-model acceleration.

3 METHODOLOGY

3.1 SPARSIFICATION INTERVENTIONS

An intervention specifies thresholding sites $\mathcal{N}$, their nonlinearities and thresholds, and pressure sites $\mathcal{P}$. The sets are independent choices, although many evaluated recipes align them (Figure 1; Appendix C.1). We use ReLU and fixed elementwise maps $G_{+, \kappa}(x) = x\mathbf{1}\{x \geq \kappa\}$ and $G_{\pm, \kappa}(x) = x\mathbf{1}\{|x| \geq \kappa\}$, with $\kappa \geq 0$. The former removes negative values and positives below κ ; the latter preserves the signs and values of surviving magnitudes. At zero threshold, their forward maps are ReLU and identity. Appendix A gives boundary derivatives. Fixed thresholds require neither ranking nor input-dependent moment estimates, but this does not establish a speed advantage over selection-based methods.

Pressure uses the mean absolute activation in each targeted site-layer tensor, after its nonlinearity, then averages these means equally. Expanding the targets changes the mixture of site gradients. *Naive LI* optimizes $\mathcal{L}_{\text{task}} + \lambda\mathcal{L}_1$, clipping the combined gradient before AdamW.

Orthogonal LI (OL1) keeps task optimization primary. After task-only AdamW, it preconditions pressure using the task’s second moments, removes the component opposing the adaptive task direction under conflict, and caps the auxiliary vector relative to that direction’s norm. This adapts gradient-conflict projection (Yu et al., 2020; Hsieh et al., 2024) while extending the norm-limiting principle of Pythia’s gradient clipping to an auxiliary update. The two caps act in different spaces.

Once this relative cap binds, increasing the pressure weight λ no longer increases the ideal correction. We use this conditional property to motivate a fixed pressure mechanism, holding budget $b = 1$

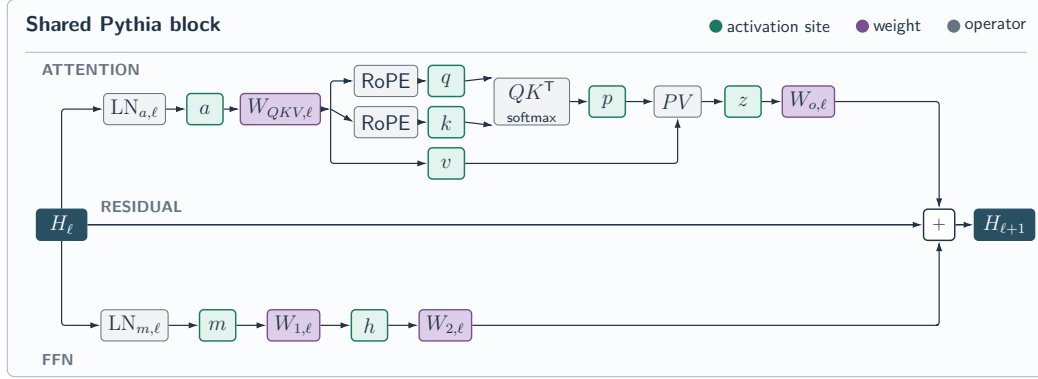


Figure 1: **Activation sites in the shared Pythia block.** A1-H replaces GELU at h with ReLU. A4 applies $G_{\pm,\kappa}$ at a, m, h, z ; A7 adds $G_{\pm,\kappa}$ at post-RoPE q, k and at v . Pressure variants target every selected site. A0 retains GELU. Appendix C.1 gives the full recipe diagram and tensor shapes.

and multisite $\lambda = 1$ while varying threshold and placement. Saturation is recipe-dependent: retained logs show frequent capping for A7-OL1 at 14M and 70M, but rarely at 410M. Appendix A.2 gives the protected-direction geometry, its distinction from task-loss preservation, and the measured cap activity.

3.2 SPARSITY AND ARCHITECTURAL REACH

Activation sparsity is the exact-zero fraction at a named site. *Model-wide activation sparsity*, $\mathcal{S}_{\text{model}} = N_0/N_{\text{model}}$, weights zeros by affected scalar multiplications. Each declared block product gets one credit when either activation operand is zero; the denominator also includes the dense LM head, which gets no zero credit. Integer counts are pooled before division, using actual operands and causally valid pairs in full, uncached sequences (Appendix B). We report $100\mathcal{S}_{\text{model}}\%$; this is multiplication opportunity, not runtime or total FLOPs.

Architectural reach, $R_{\text{arch}} = N_{\text{reach}}(\mathcal{N})/N_{\text{model}}$, describes the work structurally reached if the selected sites were entirely zero, counting overlaps once. Architecture, sites and sequence length determine it, not checkpoint quality. Natural zeros outside selected-site reach can contribute to $\mathcal{S}_{\text{model}}$, so R_{arch} need not bound every observation. Its gap to observed sparsity is not sparsity available without quality loss (Appendix B.2).

4 EXPERIMENTAL STUDY

We pretrain Pythia models from random initialization on MiniPile (Biderman et al., 2023; Kaddour, 2023): 30 conditions at 14M and 12 each at 70M and 410M. Each trains for 712 updates and 1.493 billion input tokens, with one training seed; we evaluate the final checkpoint. Loss is next-token cross-entropy (lower is better), measured with $\mathcal{S}_{\text{model}}$ over all 338 complete 2,048-token blocks from 500 validation documents; the 1,444-token tail is excluded. Appendix C.4 gives all settings, including the lower 410M learning rate.

Within each size, paired conditions share initialization, data order and budget; reused controls are not independent replications. Pressure on/off contrasts hold nonlinearities and threshold fixed. Unpressured A7 versus A4 adds symmetric q, k, v thresholding; A7-OL1 versus A4-OL1 changes complete recipes, including pressure normalization. A4 versus A1-H also changes h ’s threshold above zero and its derivative at zero. Local pressure varies $\lambda \in \{0.05, 0.1, 0.5, 1\}$; A4/A7 vary $\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$. Multisite OL1 uses $\lambda = 1$; all OL1 conditions use trust budget 1. Larger models use A0, A1-H, A4-OL1 and A7-OL1; Figure 6 lists all eight recipes.

The evaluation-only reference applies uniform magnitude clipping at a, m, h, z , calibrated separately per site/layer on ten training blocks, for targets $p = 0, 0.1, \dots, 0.9$. Target p is a quantile, not achieved $\mathcal{S}_{\text{model}}$. This TEAL-style control does not optimize allocation (Liu et al., 2024). It

Table 1: Maximum observed sparsity under same-size A0 loss plus the stated budget, across 54 trained and 540 clipped evaluations. Trained excludes A0; “none” means no eligible sparse recipe. Budgets are retrospective; identical winners are grouped. Clipped winners have no measured latency.

Size	Loss budget	Trained recipe	$\mathcal{S}_{\text{model}}$ (%)	Including clipping	$\mathcal{S}_{\text{model}}$ (%)
14M	0.05, 0.10, 0.20	A1-H-L1 $\lambda = 1$	3.949	A1-H-L1 $\lambda = 1, p = 0.3$	6.490
70M	0.05, 0.10	none	—	A0, $p = 0.2$	7.410
70M	0.20	A1-H	10.066	A1-H, $p = 0.3$	17.485
410M	0.05, 0.10	none	—	A0, $p = 0.2$	14.883
410M	0.20	A1-H	21.050	A1-H, $p = 0.3$	35.810

acts after the trained nonlinearity, preserving A0’s GELU, whereas A7 replaces GELU and also thresholds internal attention. Thus the comparison changes reach and final transformations as well as intervention timing. Appendix D.5 retains all 540 evaluations.

4.1 QUALITY–SPARSITY TRADE-OFFS

Local pressure and broader nonlinearities occupy different regimes (Figure 2). The lowest-loss local endpoint, A1-H-L1 at $\lambda = 1$, reaches 3.949% sparsity at loss 5.1023, versus A0’s 5.2086. A0 clipping at $p = 0.3$ reaches similar sparsity, 3.832%, but loss 5.3619. A7-OL1 at $\kappa = 0.5$ instead reaches 27.483% at loss 5.8294, or +0.6208 over A0. Its 29.952% architectural reach also exceeds the four-site clipping reference’s 12.833%; the comparison changes both the representation and affected computation.

Retrospective quality budgets change which endpoint is preferable (Table 1). Under a +0.05 same-size A0 loss allowance, the greatest observed sparsities are 6.490%, 7.410% and 14.883%. All use post-hoc clipping, including a pressure-trained checkpoint at 14M. No sparsified training recipe at 70M or 410M meets the two smallest budgets. These are evaluated points, without interpolation or matched latency measurements for the clipped winners.

4.2 PAIRED INTERVENTION EFFECTS

Pressure’s additional value depends on threshold and targets (Figure 3). Replacing GELU with ReLU adds 2.714 sparsity percentage points (pp) for +0.0611 loss. Local pressure lowers observed loss versus ReLU; changing naive L1 to OL1 lowers loss at three of four matched weights but reverses at $\lambda = 1$. This compares complete update rules, including clipping and optimizer state, rather than isolating conflict projection.

For A4, adding OL1 improves both axes at $\kappa = 0, 0.01$, then adds sparsity at increasing quality cost: 2.498 pp for +0.3783 loss at $\kappa = 0.5$. The corresponding A7 addition yields 12.096 pp for +0.1265; at $\kappa = 0.1$ it yields 1.372 pp for +0.0008, which does not establish loss equivalence. Without pressure, A7 minus A4 at $\kappa = 0.5$ adds 5.171 pp for +0.0432 loss. The larger pressured gain under A7 involves more targets and reweighted existing terms: it cannot be assigned solely to broader thresholding or pressure at q, k, v .

4.3 RESHAPING ACTIVATION DISTRIBUTIONS

More FFN zeros need not mean more model-wide sparsity (Figure 4). At $\kappa = 0.5$, A4-OL1 places 99.31% of pooled h, m elements at zero, versus 93.64% for A7-OL1; their attention zero fractions are 0.22% and 95.60%. A narrow nonzero density peak is therefore not a large zero probability. A4’s untargeted attention distributions change, but comparison with A0 does not isolate pressure’s contribution; A7 intervenes there directly. Appendix D.1 retains all thresholds and tails.

Operation counts explain the ranking reversal (Table 2). A7-OL1’s $QK + PV$ contribution rises by 16.72 pp while its projection contribution falls by 1.95 pp, producing greater model-wide sparsity despite fewer FFN zeros. Separate BF16 projection-input row diagnostics in Appendix D.2 do not establish residual-branch inactivity or context insensitivity: biases may remain and token identities were not retained.

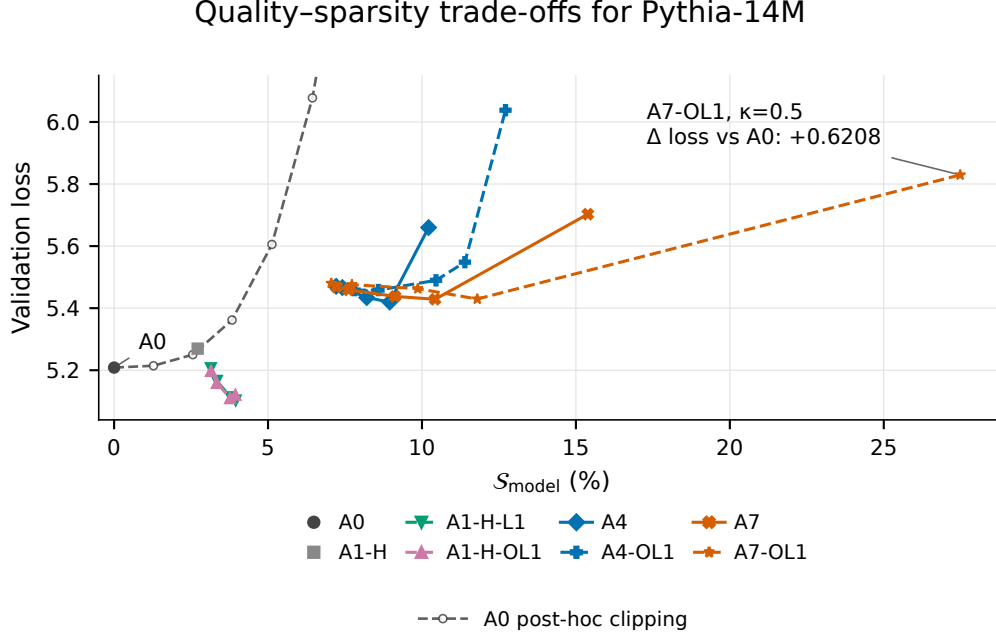


Figure 2: **Local pressure reaches lower loss; broader recipes trade quality for sparsity.** All 30 trained 14M endpoints are filled; dashed colored paths have pressure, solid paths do not. Open markers show uniform four-site A0 clipping. Six of ten clipping evaluations fall inside the displayed loss range; Appendix D.5 retains the full range. Lines join evaluated settings, not Pareto envelopes. A7 has broader reach than the clipping sites.

Table 2: 14M operation contributions at $\kappa = 0.5$. All use the full-model denominator, including the dense head with no zero credit. Projections sum QKV, attention output and both FFN projections. These are counted products, not runtime or FLOPs saved.

Recipe	Projections (pp)	$QK + PV$ (pp)	Total $S_{\text{model}} (\%)$
A4-OL1	12.657	0.057	12.713
A7-OL1	10.709	16.773	27.483

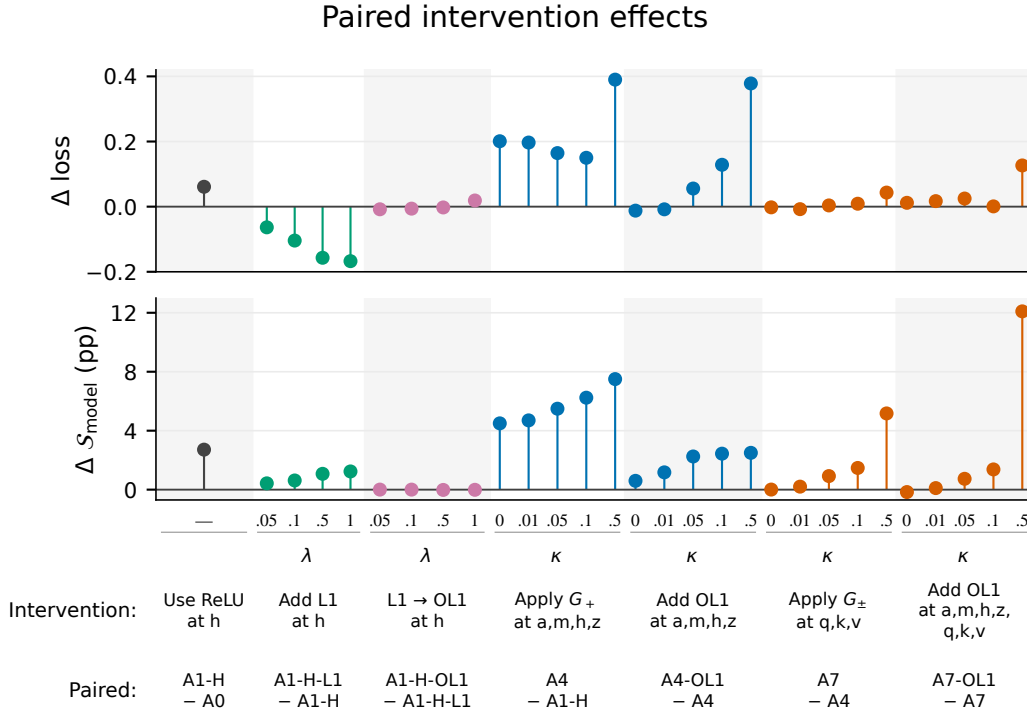


Figure 3: **Pressure’s marginal effect changes across recipes and thresholds.** All 29 paired 14M deltas are treatment minus the named reference: loss above, sparsity pp below. Labels sweep λ or κ with ordinal spacing; the dash denotes the unswept ReLU replacement. Pressure additions fix nonlinearities and threshold; other contrasts can change multiple factors (Section 4). Shared controls and one training seed do not provide 29 independent replicates. All numerical deltas are in Table 7.

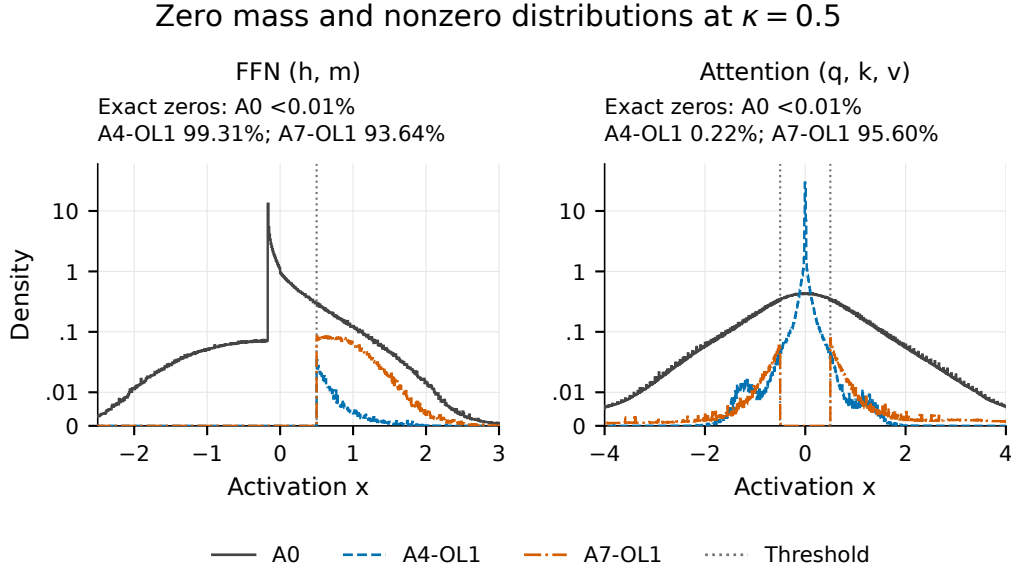


Figure 4: **A7-OL1 has fewer FFN zeros but far more attention zeros at $\kappa = 0.5$.** Exact-zero masses appear above nonzero densities. Counts pool all layers and 338 validation blocks, with $h : m$ weighting 80:20 and equal $q : k : v$ weighting, and Q/K measured after RoPE. Density divides by all elements and bin width; the symlog axis is linear below 0.01, so areas are not probabilities. Dotted lines mark thresholds. The complete grid, precision and tails are in Appendix D.1.

Table 3: A7-OL1 minus A4-OL1 at matched size and threshold. Negative loss and positive sparsity changes favor A7-OL1. These are complete-recipe contrasts; Table 8 retains all thresholds.

Size	κ	Δ loss	$\Delta \mathcal{S}_{\text{model}}$ (pp)
14M	0	+0.0219	-0.756
14M	0.5	-0.2086	+14.769
70M	0	+0.1358	-2.110
70M	0.5	-0.1736	+5.006
410M	0	-0.2658	+2.783
410M	0.5	-0.0703	+9.024

4.4 TRANSFER ACROSS MODEL SIZES

At $\kappa = 0.5$, A7-OL1 improves both loss and model-wide sparsity over A4-OL1 at all three sizes (Table 3). At zero threshold, A4-OL1 is better at 14M and 70M, while A7-OL1 is better at 410M. This transfers a complete-recipe relationship, not every threshold effect: larger cohorts lack no-pressure A4/A7 controls, share tokens rather than tokens per parameter, and use a lower learning rate at 410M.

A7 reach rises from 29.95% to 49.42% and 87.25% as the dense head’s share of counted work falls. Figure 8 therefore also reports block-only sparsity $\mathcal{S}_{\text{block}} = \mathcal{S}_{\text{model}}/R_{\text{arch}}(\text{A7})$, using the same A7 reference for every curve within a size. This removes head share but retains differing block-operation weights; it is not a quality scaling law. Trained endpoints improve on uniform A0 clipping at aggressive $p \geq 0.5$ targets at all sizes, including a check beyond observed $p = 0$ numerical drift (Appendix D.5).

4.5 REALIZING SPARSITY WITH SPECIALIZED INFERENCE KERNELS

Released TwELL kernels accelerated their supported workload, but the unchanged small-width primitive failed Pythia-14M qualification. Its adapted P0 descendant qualifies on only four of our 30 checkpoints; the 0.8910 $\times$ subset mean is not comparable to a full-cohort mean (Appendix D.6).

We use a human-guided coding-agent loop and replay 42 eligible proposals against native PyTorch/SDPA CUDA graphs. The workload is one RTX5090, BF16, batch one, uncached 2,048-token inference with full vocabulary logits. Qualification checks logits and loss within specified tolerances on all 338 validation blocks. The recorded agent default is *gpt-6-astra*; per-request served models and superiority over human-only development are unverified.

Fusion gains and conditional benefit from sparse execution

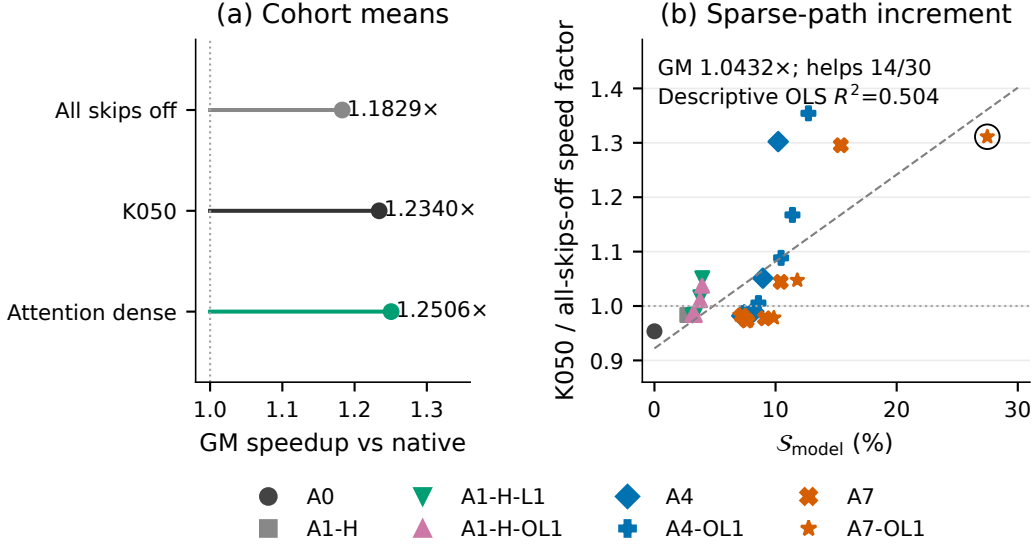


Figure 5: **Fusion gains and a conditional contribution from sparse execution.** (a) Equal-checkpoint geometric mean native-relative speedups over all 30 14M checkpoints. (b) K050’s speedup divided by all-skips-off’s; values above one favor sparse paths. The circle marks the search checkpoint; the dashed line is descriptive OLS. Timings use the BF16 workload above, while the horizontal axis uses canonical FP16 counts. Axes are restricted; neither the fit nor same-checkpoint ratios establish a quality-matched model advantage.

Selected K050 qualifies on all 30 checkpoints, with geometric mean native-relative speedup $1.2340\times$. Its search retrospective reaches $1.7830\times$ at the fixed A7-OL1, $\kappa = 0.5$ checkpoint; Appendix D.6 retains the history and distinguishes the final-sweep maximum. Fusion alone gives $1.1829\times$ (Figure 5). Relative to that matched all-skips-off control, sparse paths add $1.311\times$ at the search checkpoint and $1.043\times$ geometrically, helping 14 of 30 checkpoints. These factors derive from separately measured native-normalized timings. Their association with canonical FP16 sparsity is weaker than native-relative speedup’s ($R^2 = 0.5037$ versus 0.8167); Appendix D.6.1 retains all descriptive sensitivity checks and absolute latencies. BF16 timing and FP16 counts remain distinct.

Disabling attention skipping improves every checkpoint, raising the mean to $1.2506\times$. The kernel skips an instruction when either complete operand fragment is zero. It bypasses 58.0% of eligible *QK* and 67.2% of *PV* instructions at the search checkpoint, yet runs slower than the dense attention ablation. Partial zeros can still require an instruction, while detection and control cost time. Scalar zero opportunity alone therefore does not identify profitable execution.

5 DISCUSSION AND CONCLUSION

Pressure’s additional value depends on the threshold and target objective: it can improve both observed axes at one setting and exchange quality for zeros at another. Broader attention intervention can also raise model-wide sparsity while reducing pooled FFN sparsity. Recipe comparisons therefore require paired quality changes and operation-weighted counts. Selecting a model under

a loss budget and selecting its execution kernel require further comparisons: neither a high sparsity percentage nor a same-checkpoint native-relative speedup establishes a quality-matched model advantage.

Architectural reach, observed zero products, skipped instructions and runtime are distinct quantities. Here, fusion and sparse paths both contribute, but attention skipping loses despite bypassing many instructions. These findings concern one-seed MiniPile pretraining, selected larger-size recipes, a limited uniform-clipping reference and one 14M kernel workload, with FP16 counts and BF16 timing. The next questions are whether the pressure relationships replicate with existing coefficients held fixed, and whether the execution trade-off persists at larger widths.

AI USE STATEMENT

AI assistance supported code for the reported reductions, inspection of experiment records, interpretation of results, literature checks, preparation of figures and tables, and drafting and editing this manuscript. The kernel case study additionally used a human-guided coding-agent loop for implementation, testing and optimization. Its recorded default model was *gpt-6-astra*; per-request served-model identities were not verified. Reported measurements are tied to retained execution records. Numerical reductions were checked against source counts and timing samples, and kernel claims use the qualification procedure in Appendix D.6.

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A INTERVENTION DEFINITIONS

A.1 THRESHOLDING NONLINEARITIES

For finite $\kappa \geq 0$, both thresholding nonlinearities retain equality and leave retained values unchanged. The comparison mask is detached: the derivative with respect to the input is one for retained entries and zero for rejected entries. Consequently, $G_{+,0}$ and ReLU have identical forward values but differ at zero, where their derivatives are one and zero, respectively. $G_{\pm,0}$ is exactly the identity in both values and gradients. These nonlinearities use neither a straight-through estimator nor a top- k selection rule. Unselected sites retain their original transformations.

A.2 ACTIVATION PRESSURE AND OL1

For the J targeted site-layer tensors A_j , containing n_j entries each, the pressure objective is $\mathcal{L}_1 = J^{-1} \sum_{j=1}^J n_j^{-1} \sum_{i=1}^{n_j} |A_{j,i}|$. The tensors contain the executed site activations, after the nonlinearity where one is applied. Absolute values and means are evaluated in float32. Each tensor receives equal weight regardless of its size. Adding pressure sites changes the aggregate direction and, when the cap is inactive, its scale. Task and pressure losses are averaged over the same accumulation microbatches. For naive L1, the accumulated gradient of $\mathcal{L}_{\text{task}} + \lambda \mathcal{L}_1$ is globally norm-clipped before AdamW (Loshchilov & Hutter, 2019).

For OL1, let g_t and g_1 be the accumulated task and unweighted pressure gradients. We first clip g_t by its global norm and perform AdamW using only that clipped task gradient. Let $\hat{m}_t$ and $\hat{v}_t$ be its newly updated, bias-corrected moments. On parameters having both gradients and initialized optimizer state, define the adaptive task and pressure vectors $u = \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon_A)$ and $w = g_1 / (\sqrt{\hat{v}_t} + \epsilon_A)$. The pressure gradient is unclipped and has no first-moment buffer. Inner products and norms below pool all these eligible parameters. With $c = \langle u, w \rangle$, $q = \|u\|_2^2$, and numerical stabilizer $\epsilon > 0$, the conflict-conditioned projection is

$$\tilde{w} = \begin{cases} w - \frac{c}{q + \epsilon} u, & c < 0 \text{ and } q > \epsilon, \\ w, & \text{otherwise.} \end{cases} \quad (1)$$

For pressure weight $\lambda \geq 0$ and required budget $b > 0$, define the norm ratio $r = \lambda \|\tilde{w}\|_2 / (\|u\|_2 + \epsilon)$. Set $s = \min\{1, b/(r + \epsilon)\}$ when $r > 0$, or $s = 1$ otherwise. After AdamW, each eligible parameter group p , with learning rate α_p , receives $\theta_p \leftarrow \theta_p - \alpha_p \lambda s \tilde{w}_p$. The experiments' clipping thresholds, b , and λ are condition parameters.

The executed FP16 boundaries accumulate and unscale the two gradients separately. A nonfinite gradient skips the entire boundary, including the correction. Missing gradients or uninitialized moment state exclude a parameter from its pooled geometry. All reported OL1 runs use $b = 1$ and $\epsilon = 10^{-12}$; the local study sweeps $\lambda \in \{0.05, 0.1, 0.5, 1\}$, whereas multisite runs fix $\lambda = 1$. The norms precede group learning rates and exclude AdamW's decoupled weight decay.

Protected direction and saturation. For the ideal rule, omit stabilizers and take $u \neq 0$. Project only under conflict, using $\tilde{w} = w - (u^\top w / \|u\|^2) u$; otherwise retain w . Then the pressure descent vector is

$$d_p = \min\left(\lambda, \frac{b\|u\|}{\|\tilde{w}\|}\right) \tilde{w}, \quad d_p = 0 \text{ if } \tilde{w} = 0. \quad (2)$$

It satisfies $u^\top d_p \geq 0$, with equality after a conflicting projection, and hence $u^\top (u + d_p) \geq \|u\|^2$. Also $\|d_p\| \leq b\|u\|$: the auxiliary vector is capped, while the combined vector can be longer than u . At a fixed gradient pair and optimizer state, $\lambda \|\tilde{w}\| \geq b\|u\|$ makes the vector independent of further increases in λ . A common learning rate α turns this into the added displacement $-\alpha d_p$.

With $\rho = b\|u\|$ and nonzero $\tilde{w}$, the saturated vector maximizes $w^\top d$ subject to $u^\top d \geq 0$ and $\|d\| \leq \rho$. Without conflict this follows from Cauchy-Schwarz. Under conflict, $w = \tilde{w} + cu/\|u\|^2$ with $c < 0$, so every feasible d satisfies $w^\top d \leq \tilde{w}^\top d \leq \rho \|\tilde{w}\|$; the stated vector attains both bounds. If w exactly opposes u , zero is an optimum, possibly nonunique. This is alignment with preconditioned pressure, not necessarily the first-order decrease $g_1^\top d$.

The protected adaptive direction generally differs from the current task gradient: first-order task loss depends on $g_t^\top d_p$, and finite steps also depend on curvature. Thus the property does not guarantee task-loss preservation. In the implementation, stabilizers make the projection approximately orthogonal, and $q \leq \epsilon$ bypasses it altogether.

Observed cap activity and objective scaling. All 34 OL1 conditions retain all 712 boundary records, with no skipped updates. The cap is active exactly when $r > 0$ and $b/(r + \epsilon) < 1$, using the logged pre-learning-rate ratio on eligible parameters. It binds on 7,283 of 24,208 boundaries (30.08%), but this pooled fraction hides large recipe differences. Across thresholds, A7-OL1 binds on 84.97–100% of steps at 14M and 100% at 70M, versus 0.42–0.84% at 410M. A4-OL1 binds on 0.14–2.67% across all sizes; local h-only OL1 ranges from 0 to 10.67% across its four weights. The first boundary has zero learning rate: its cap diagnostic describes a vector, not parameter movement. Supplementary records retain counts, correction-ratio distributions and separate early (steps 1–142) and late (570–712) summaries for every condition.

Multiplying a fixed objective by a positive scalar preserves the saturated ideal vector’s direction and norm when both versions remain saturated. For a fixed seven-site target set, replacing $\sum_7/(7L)$ by $\sum_7/(4L)$ only rescales that vector before the cap. Expanding four targets to seven instead changes their gradient mixture, even under saturation. Smaller coefficients in the written mean therefore do not establish proportional dilution of the effective update. Relative weights between old and new targets would test a different direction; the current experiments do not identify that contribution separately.

B SPARSITY ACCOUNTING

B.1 EXACT COUNTERS

Activation sparsity at site s is $\mathcal{Z}_s = \sum_{\ell,e,i} \mathbf{1}\{A_{s,i}^{(\ell,e)} = 0\} / \sum_{\ell,e} |A_s^{(\ell,e)}|$, where ℓ indexes layers, e evaluation batches, i tensor entries, and $|A|$ the number of elements in tensor A . Near-zero mass replaces equality by $|A_{s,i}| \leq \varepsilon$ for a stated tolerance; it does not contribute to exact-zero counts unless the value is zero. All aggregate fractions pool integer numerators and denominators across batches and layers before division.

The atomic unit of model-wide accounting is one scalar multiplication in a matrix product. If $X \in \mathbb{R}^{n \times p}$ feeds a linear projection with q outputs, the count is $N = npq$, with $N_0 = q \sum_{i,j} \mathbf{1}\{X_{ij} = 0\}$. Zero weights receive no credit. For the multihead attention graph specified in Appendix C.1, both operands are activations, so a product is credited once if either is zero. Within one block and evaluation batch, for query position t , key position $u \leq t$, and within-head coordinate j ,

$$N_{0,QK} = \sum_{b,h,t} \sum_{u \leq t} \sum_j \mathbf{1}\{Q_{bhtj} = 0 \vee K_{bhuj} = 0\}, \quad (3)$$

$$N_{0,PV} = \sum_{b,h,t} \sum_{u \leq t} \sum_j \mathbf{1}\{P_{bhtu} = 0 \vee V_{bhuj} = 0\}. \quad (4)$$

Here P is the causal softmax probability tensor; the model numerator sums these counts over blocks and evaluation batches. Future masked positions are excluded from both counts, while zeros on valid positions, including floating-point underflow, are included. Counts use the operands actually consumed by each matrix product; upstream masks do not determine those operands’ zeros after intervening transformations. Earlier value positions participate in more causal pairs, so local value sparsity alone does not determine PV sparsity.

B.2 ARCHITECTURAL REACH AND NORMALIZATION

For a specified graph and workload, let N_o count the valid scalar multiplications in block operation family o , pooled over layers and examples. Then $N_{\text{model}} = \sum_o N_o + N_{\text{head}}$, while N_0 sums the corresponding zero-operand counts only in block operations. The block-only view is $\mathcal{S}_{\text{block}} = N_0 / \sum_o N_o$. Both sparsities are fractions; percentages multiply them by 100. Embedding lookup, normalization, nonlinearities, softmax, RoPE, bias and residual additions, and data movement are outside this accounting.

To construct the reach numerator, count products guaranteed to have a zero activation operand as a structural consequence of 100% activation sparsity at the selected sites, independently of checkpoint values or natural zeros elsewhere. Include exact zero-preserving propagation under the declared graph rules. Products reached through more than one site receive one credit. The resulting count is $N_{\text{reach}}(\mathcal{N})$. Division by the same N_{model} gives R_{arch} . Natural zeros outside this selected-site reach can contribute to observed $\mathcal{S}_{\text{model}}$, so architectural reach is not an upper bound on every observation. Its gap to an observation does not measure sparsity available without quality loss. Appendix C.2 specializes the denominator and reach rules to the Pythia graph used in the experiments.

C EXPERIMENTAL SETUP DETAILS

C.1 PYTHIA SITES AND NONLINEARITIES

Pythia uses parallel attention and FFN branches with separate LayerNorms. For batch size B , sequence length T , residual width d , FFN width d_f , and H heads, let $d_h = d/H$. Table 4 identifies the actual operands recorded after the selected nonlinearity. The same site mapping is applied in every layer, with one common threshold per condition. Figure 1 shows the shared block; Figure 6 below retains all recipe, pressure and reach assignments.

Table 4: Intervention sites and the matrix products that consume them directly.

Site	Shape	Consuming operation	Placement
a	$B \times T \times d$	QKV projection	After attention LayerNorm
m	$B \times T \times d$	FFN up projection	After FFN LayerNorm
h	$B \times T \times d_f$	FFN down projection	Nonlinearity output
q, k	$B \times H \times T \times d_h$	Attention scores	After partial RoPE
v	$B \times H \times T \times d_h$	Probability-value	After the QKV split
z	$B \times T \times d$	Attention output	Concatenated context, before output projection

At h , the selected nonlinearity replaces GELU. Other sites retain their original activations unless a thresholding nonlinearity is applied. At zero threshold, A7’s additional symmetric query/key/value nonlinearities are exact identities in both values and gradients. They therefore do not change A4’s mathematical forward map in the absence of pressure. A7-OL1 still differs from A4-OL1 in its pressure targets and their equal-tensor normalization.

Query and key counts use actual post-RoPE operands: rotation can repopulate an individual zero coordinate. Likewise, the output projection uses actual post-nonlinearity context z , rather than inferring its zeros from the value mask.

C.2 ARCHITECTURE-SPECIFIC COUNTS AND ARCHITECTURAL REACH

The operation inventory comprises QKV projection, attention scores (QK), probability-value products (PV), attention-output projection, and the two FFN projections. For one uncached, unpadded sequence, each block contributes $3Td^2$ products from QKV projection, Td^2 from the attention-output projection, Tdd_f from each FFN projection, and $dT(T+1)/2$ from each of QK and PV . Thus, for L blocks and vocabulary size V_{vocab} ,

$$\begin{aligned} C_{\text{block}} &= T(4d^2 + 2dd_f) + dT(T+1), \\ C_{\text{model}} &= LC_{\text{block}} + TdV_{\text{vocab}}. \end{aligned} \tag{5}$$

Over B sequences, $N_{\text{model}} = BC_{\text{model}}$; the dense head contributes only to the denominator, as defined in Section 3.2.

Let $\mathcal{O}(\mathcal{N})$ be the union of operations reachable when the selected sites have 100% activation sparsity: a reaches QKV; m reaches FFN up; h reaches FFN down; either q or k reaches QK ; v reaches PV and attention output; and z reaches attention output. Each operation is counted once. The v rule includes the exact closure $V = 0 \Rightarrow PV = 0$. No further propagation is assumed: QKV biases can survive $a = 0$, and zero scores do not imply zero softmax probabilities. For the same site set in

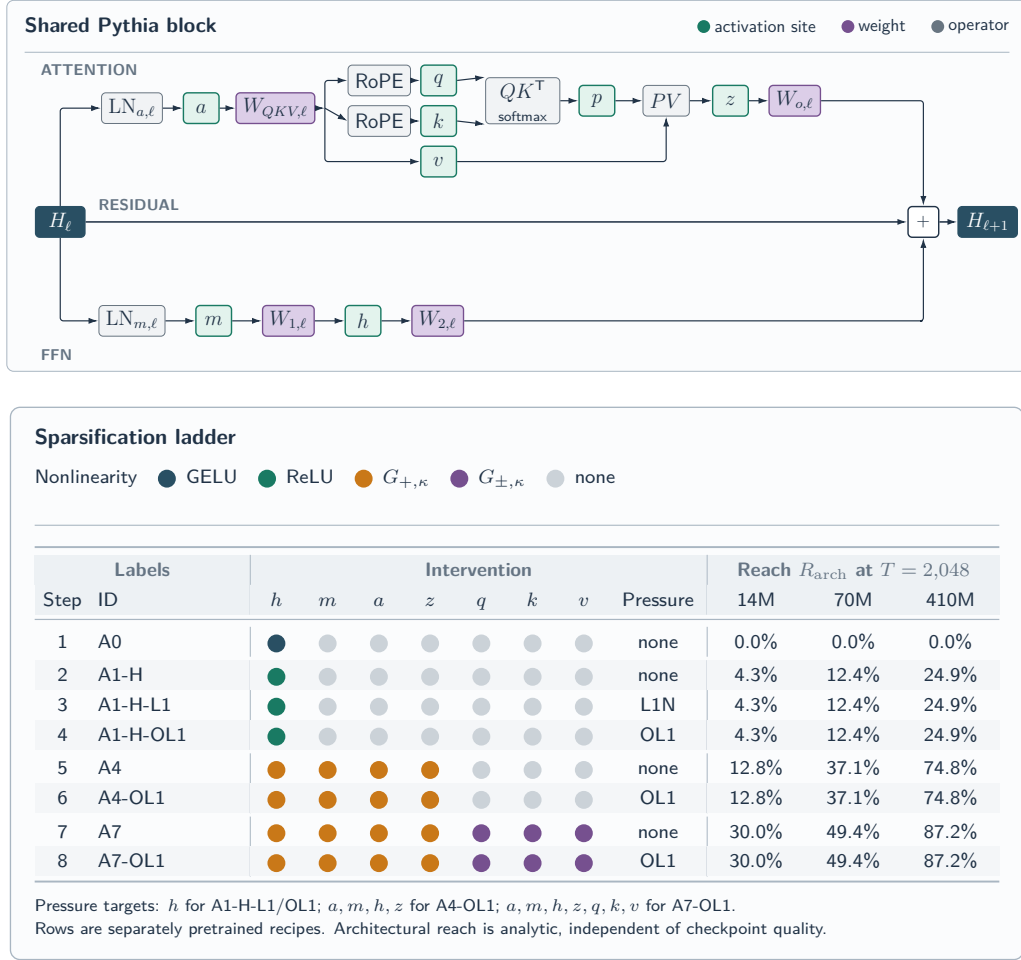


Figure 6: **Pythia architecture and sparsification recipes.** Top: the shared block and activation sites; q, k follow RoPE and z precedes the attention-output projection. Bottom: nonlinearity placement, pressure and architectural reach for the eight recipes. Selected nonlinearities replace GELU at h and act on existing activations elsewhere. L1N denotes naive L1; pressure targets every site with a selected nonlinearity in its row. The reach columns report $100R_{\text{arch}}\%$ for $T = 2,048$, rather than measured sparsity or speedup. All recipes are evaluated at 14M; 70M and 410M use A0, A1-H, A4-OL1 and A7-OL1. Rows identify separately trained recipes, not curriculum stages.

every layer, $N_{\text{reach}} = BL \sum_{o \in \mathcal{O}(\mathcal{N})} C_o$, where C_o is that operation’s per-sequence count. Division by N_{model} gives R_{arch} .

With Pythia’s $d_f = 4d$, the per-block reachable counts for A0, A1-H, A4, and A7 are 0, $4Td^2$, $12Td^2$, and C_{block} , respectively. The ladder evaluates these formulas at $T = 2,048$, $V_{\text{vocab}} = 50,304$, and the pinned (L, d) configurations (6, 128), (6, 512), and (24, 1,024).¹ No checkpoint enters this calculation. For example, A0 has zero selected-site reach but may have positive observed sparsity from natural zeros.

C.3 EVALUATION SCOPE

Quality and logical counts are paired in the same eager-attention evaluation of a checkpoint, with its declared nonlinearities and evaluation precision. The complete MiniPile validation split contains 500 documents. Concatenation and packing produce 338 complete blocks of 2,048 tokens (692,224 input tokens); the final 1,444-token tail is excluded. Each block contributes 2,047 next-token targets to cross-entropy, while product accounting includes all 2,048 input positions. Loss is averaged over the equally sized blocks, and sparsity is count-pooled. This workload uses no key-value cache. Condition-specific training budgets, optimizer settings, seeds, and checkpoint selection accompany their results; they are not implied by the ladder’s list of candidate conditions.

C.4 REALIZED TRAINING PROTOCOL

We use MiniPile and the Pythia-14M-deduped tokenizer for every size. Each document is encoded without added special tokens, followed by one end-of-sequence token. Concatenated tokens are packed into nonoverlapping 2,048-token blocks. Training uses a seeded permutation of the block indices; validation retains source order. The training split contains 1,000,000 documents and 1,491,711,416 tokens, yielding 728,374 complete blocks and an excluded 1,464-token tail. The supplementary `protocol.json` records immutable dataset and tokenizer revisions, cache hashes and implementation identities.

All conditions start from random weights drawn from the pinned Pythia architecture configurations; released pretrained weights are not used. Model initialization and data-order seeds are both 1234, with one training seed per condition. Parameter sweeps measure intervention variability rather than variation over independent training seeds. All quality comparisons reuse the same validation split, without a separate confirmation set. Within a size, the conditions share the initial parameter hash and training schedule. The 30-condition 14M cohort contains A0, A1-H, four weights each for A1-H-L1 and A1-H-OL1, and five thresholds each for A4, A4-OL1, A7 and A7-OL1. Each larger cohort contains A0, A1-H, and the five-threshold A4-OL1/A7-OL1 sweeps. Local pressure weights are $\{0.05, 0.1, 0.5, 1\}$; thresholds are $\{0, 0.01, 0.05, 0.1, 0.5\}$. Multisite OL1 uses $\lambda = 1$, with pressure at all four or seven selected sites. Every OL1 condition uses trust budget 1 and projection stabilizer $\epsilon = 10^{-12}$.

Training uses AdamW with $\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$, weight decay 0.1 and global task-gradient clipping at norm 1. Naive L1 clips the combined objective gradient instead. The learning rate warms up linearly for eight updates and follows cosine decay to one tenth of its peak. Parameters and optimizer state are FP32; forward computation uses FP16 autocast with dynamic loss scaling. Attention and hidden dropout are zero. Table 5 gives size-dependent settings. The global batch is 1,024 sequences of 2,048 tokens, for 712 optimizer updates and 1,493,172,224 input tokens. This wraps 714 blocks beyond the 728,374 complete blocks in one packed training pass. We use the final checkpoint without validation-based checkpoint selection. The lower 410M learning rate is part of the realized protocol, so size alone is not the only cross-scale change.

¹Architecture configurations: EleutherAI’s Pythia-14M-deduped, Pythia-70M-deduped, and Pythia-410M-deduped. Links identify immutable configuration revisions; released weights are not used.

Table 5: Size-dependent training settings. Tokens per parameter use the common realized input-token budget; they are not held fixed across sizes.

Setting	14M	70M	410M
Parameters	14,067,712	70,426,624	405,334,016
Peak learning rate	10^{-3}	10^{-3}	3×10^{-4}
Minimum learning rate	10^{-4}	10^{-4}	3×10^{-5}
Microbatch (sequences)	32	4	4
Gradient accumulation	32	256	256
Tokens per parameter	106.14	21.20	3.68

D COMPLETE RESULTS AND SUPPORTING DIAGNOSTICS

The following tables report every selected trained endpoint, all 29 paired 14M contrasts and all 15 cross-scale recipe comparisons. They retain dominated settings as well as favorable ones. Each entry summarizes one trained model on complete validation. Sparsity percentages are formed from pooled integer product counts. The endpoint table uses the common A7 architectural-reach denominator from Figure 8, rather than a different denominator for each recipe. This common normalization equals S_{block} .

D.1 ACTIVATION SPARSITY AND SIGNED-DENSITY COVERAGE

The seven frozen-checkpoint evaluations underlying Figure 7 record signed values after the trained nonlinearities, with Q/K measured after RoPE. Each checkpoint contributes 2,658,140,160 FFN elements and 1,594,884,096 attention elements across the complete validation set. Summing integer counts gives the 80:20 weighting of $h : m$ and equal weighting of $q : k : v$. These pooled distributions are element-weighted mixtures, not distributions of layer means.

Exact zeros are stored separately from the nonzero histogram. The native grid spans $[-8, 8]$ with 16,000 bins of nominal width 0.001; the figure combines ten bins at a time and divides by the actual floating-point bin width and the count of all captured elements. Consequently, a curve’s full integral is the recorded nonzero mass inside that grid, not one. Table 9 reports the zero mass and the fractions outside the displayed and native ranges. Underflow and overflow counts are retained rather than folded into edge bins. The displayed limits are $[-2.5, 3]$ for FFN and $[-4, 4]$ for attention. The largest omitted display tail is 6.6719%; the native-grid tail is at most 0.7607%.

The diagnostic reruns use FP32 parameters with FP16 autocast and no additional clipping. Their maximum absolute validation-loss difference from the retained reference is 6.89×10^{-5} , below the specified 5×10^{-4} tolerance; all 108 threshold-consistency checks pass. Small rounding differences in exact-zero counts between evaluation passes do not change the displayed conclusions. Complete per-site and pooled histograms, counts and checkpoint hashes accompany the data release.

D.2 PROJECTION-INPUT VECTORS IN THE BF16 KERNEL DIAGNOSTIC

An untimed diagnostic pass retained the distribution of nonzero counts per token row at each site and layer for all 30 K050 checkpoints. These are BF16 kernel operands on all 338 validation blocks, separate from the FP16 histogram reruns above. Every layer contributes 692,224 rows; h has 512 coordinates per row and z has 128. Table 10 reports the two $\kappa = 0.5$ pressured recipes. Summing the row histograms recovers the captured sitewise zero counts. Pooling rows across layers gives the all-zero-vector rates reported below. The records do not retain token identities or context perturbations, and zero projection inputs can still produce bias outputs. They therefore do not establish whether a branch has lost its function.

At $\kappa = 0.5$, pooling all layers gives all-zero h/z row fractions of 83.72%/97.15% for A4-OL1 and 63.12%/91.20% for A7-OL1.

Table 6: All 54 trained endpoints. Δ loss uses same-size A0. λ is pressure weight; κ is the trained threshold. Block-only sparsity $\mathcal{S}_{\text{block}} = \mathcal{S}_{\text{model}}/R_{\text{arch}}(\text{A7})$ uses the same A7 reference within each size, including A0. Rounded zero sparsity need not be exactly zero.

Size	Recipe	Parameter	Loss	Δ loss	$\mathcal{S}_{\text{model}}$ (%)	$\mathcal{S}_{\text{block}}$ (%)
14M	A0	—	5.2086	+0.0000	0.000	0.000
14M	A1-H	—	5.2696	+0.0611	2.714	9.061
14M	A1-H-L1	$\lambda = 0.05$	5.2061	-0.0024	3.142	10.489
14M	A1-H-L1	$\lambda = 0.1$	5.1655	-0.0431	3.336	11.139
14M	A1-H-L1	$\lambda = 0.5$	5.1127	-0.0959	3.788	12.645
14M	A1-H-L1	$\lambda = 1$	5.1023	-0.1063	3.949	13.185
14M	A1-H-OL1	$\lambda = 0.05$	5.1981	-0.0105	3.145	10.501
14M	A1-H-OL1	$\lambda = 0.1$	5.1594	-0.0492	3.339	11.146
14M	A1-H-OL1	$\lambda = 0.5$	5.1102	-0.0983	3.768	12.580
14M	A1-H-OL1	$\lambda = 1$	5.1212	-0.0874	3.938	13.149
14M	A4	$\kappa = 0$	5.4705	+0.2619	7.212	24.078
14M	A4	$\kappa = 0.01$	5.4665	+0.2579	7.414	24.752
14M	A4	$\kappa = 0.05$	5.4341	+0.2255	8.206	27.396
14M	A4	$\kappa = 0.1$	5.4197	+0.2111	8.953	29.891
14M	A4	$\kappa = 0.5$	5.6597	+0.4511	10.216	34.106
14M	A4-OL1	$\kappa = 0$	5.4583	+0.2497	7.810	26.075
14M	A4-OL1	$\kappa = 0.01$	5.4583	+0.2497	8.587	28.668
14M	A4-OL1	$\kappa = 0.05$	5.4898	+0.2812	10.456	34.910
14M	A4-OL1	$\kappa = 0.1$	5.5483	+0.3397	11.394	38.041
14M	A4-OL1	$\kappa = 0.5$	6.0380	+0.8294	12.713	42.446
14M	A7	$\kappa = 0$	5.4684	+0.2598	7.218	24.097
14M	A7	$\kappa = 0.01$	5.4588	+0.2502	7.618	25.433
14M	A7	$\kappa = 0.05$	5.4379	+0.2293	9.127	30.471
14M	A7	$\kappa = 0.1$	5.4287	+0.2201	10.425	34.805
14M	A7	$\kappa = 0.5$	5.7029	+0.4943	15.387	51.371
14M	A7-OL1	$\kappa = 0$	5.4802	+0.2716	7.054	23.551
14M	A7-OL1	$\kappa = 0.01$	5.4758	+0.2672	7.723	25.786
14M	A7-OL1	$\kappa = 0.05$	5.4628	+0.2542	9.863	32.929
14M	A7-OL1	$\kappa = 0.1$	5.4295	+0.2209	11.797	39.385
14M	A7-OL1	$\kappa = 0.5$	5.8294	+0.6208	27.483	91.755
70M	A0	—	4.0998	+0.0000	0.000	0.001
70M	A1-H	—	4.2227	+0.1230	10.066	20.366
70M	A4-OL1	$\kappa = 0$	4.8054	+0.7056	25.673	51.944
70M	A4-OL1	$\kappa = 0.01$	4.8229	+0.7231	26.224	53.060
70M	A4-OL1	$\kappa = 0.05$	4.8742	+0.7744	28.490	57.644
70M	A4-OL1	$\kappa = 0.1$	4.9439	+0.8441	30.574	61.861
70M	A4-OL1	$\kappa = 0.5$	5.3895	+1.2897	35.596	72.022
70M	A7-OL1	$\kappa = 0$	4.9412	+0.8414	23.562	47.674
70M	A7-OL1	$\kappa = 0.01$	4.9364	+0.8366	24.226	49.017
70M	A7-OL1	$\kappa = 0.05$	4.9637	+0.8639	26.524	53.666
70M	A7-OL1	$\kappa = 0.1$	4.9501	+0.8503	28.694	58.056
70M	A7-OL1	$\kappa = 0.5$	5.2159	+1.1162	40.602	82.150
410M	A0	—	4.5475	+0.0000	0.011	0.013
410M	A1-H	—	4.6513	+0.1038	21.050	24.127
410M	A4-OL1	$\kappa = 0$	5.6921	+1.1446	38.339	43.944
410M	A4-OL1	$\kappa = 0.01$	5.6914	+1.1439	39.149	44.873
410M	A4-OL1	$\kappa = 0.05$	5.6789	+1.1315	42.125	48.283
410M	A4-OL1	$\kappa = 0.1$	5.6789	+1.1314	45.688	52.367
410M	A4-OL1	$\kappa = 0.5$	5.1910	+0.6435	71.591	82.058
410M	A7-OL1	$\kappa = 0$	5.4263	+0.8788	41.121	47.133
410M	A7-OL1	$\kappa = 0.01$	5.4331	+0.8857	41.478	47.542
410M	A7-OL1	$\kappa = 0.05$	5.4679	+0.9204	44.053	50.494
410M	A7-OL1	$\kappa = 0.1$	5.4907	+0.9433	48.998	56.161
410M	A7-OL1	$\kappa = 0.5$	5.1207	+0.5732	80.616	92.401

Table 7: All 29 Pythia-14M contrasts, treatment minus reference. When both recipes have a swept parameter, its value is matched; A0 and A1-H are fixed controls.

Treatment – reference	Parameter	Δ loss	$\Delta\mathcal{S}_{\text{model}}$ (pp)
A1-H – A0	—	+0.0611	+2.714
A1-H-L1 – A1-H	$\lambda = 0.05$	-0.0635	+0.427
A1-H-L1 – A1-H	$\lambda = 0.1$	-0.1042	+0.622
A1-H-L1 – A1-H	$\lambda = 0.5$	-0.1569	+1.073
A1-H-L1 – A1-H	$\lambda = 1$	-0.1674	+1.235
A1-H-OL1 – A1-H-L1	$\lambda = 0.05$	-0.0081	+0.004
A1-H-OL1 – A1-H-L1	$\lambda = 0.1$	-0.0061	+0.002
A1-H-OL1 – A1-H-L1	$\lambda = 0.5$	-0.0025	-0.020
A1-H-OL1 – A1-H-L1	$\lambda = 1$	+0.0189	-0.011
A4 – A1-H	$\kappa = 0$	+0.2009	+4.498
A4 – A1-H	$\kappa = 0.01$	+0.1969	+4.700
A4 – A1-H	$\kappa = 0.05$	+0.1645	+5.492
A4 – A1-H	$\kappa = 0.1$	+0.1500	+6.239
A4 – A1-H	$\kappa = 0.5$	+0.3900	+7.501
A4-OL1 – A4	$\kappa = 0$	-0.0122	+0.598
A4-OL1 – A4	$\kappa = 0.01$	-0.0082	+1.173
A4-OL1 – A4	$\kappa = 0.05$	+0.0556	+2.251
A4-OL1 – A4	$\kappa = 0.1$	+0.1286	+2.441
A4-OL1 – A4	$\kappa = 0.5$	+0.3783	+2.498
A7 – A4	$\kappa = 0$	-0.0021	+0.006
A7 – A4	$\kappa = 0.01$	-0.0077	+0.204
A7 – A4	$\kappa = 0.05$	+0.0038	+0.921
A7 – A4	$\kappa = 0.1$	+0.0090	+1.472
A7 – A4	$\kappa = 0.5$	+0.0432	+5.171
A7-OL1 – A7	$\kappa = 0$	+0.0118	-0.163
A7-OL1 – A7	$\kappa = 0.01$	+0.0170	+0.106
A7-OL1 – A7	$\kappa = 0.05$	+0.0249	+0.736
A7-OL1 – A7	$\kappa = 0.1$	+0.0008	+1.372
A7-OL1 – A7	$\kappa = 0.5$	+0.1265	+12.096

Table 8: All matched A7-OL1 minus A4-OL1 comparisons. Negative loss changes and positive sparsity changes favor A7-OL1.

Size	κ	Δ loss	$\Delta\mathcal{S}_{\text{model}}$ (pp)
14M	0	+0.0219	-0.756
14M	0.01	+0.0175	-0.863
14M	0.05	-0.0270	-0.593
14M	0.1	-0.1188	+0.402
14M	0.5	-0.2086	+14.769
70M	0	+0.1358	-2.110
70M	0.01	+0.1135	-1.998
70M	0.05	+0.0895	-1.966
70M	0.1	+0.0062	-1.880
70M	0.5	-0.1736	+5.006
410M	0	-0.2658	+2.783
410M	0.01	-0.2582	+2.329
410M	0.05	-0.2111	+1.929
410M	0.1	-0.1881	+3.310
410M	0.5	-0.0703	+9.024

Table 9: Activation sparsity (the probability mass at zero) omitted from the density curves, and mass outside their displayed range (view tail) and the native $[-8, 8]$ histogram (grid tail). Grid tails are a subset of view tails, not an additional mass to add. All percentages divide integer counts by all captured elements. A0 is listed once although it is repeated in each figure row. These 14 groups come from seven checkpoints.

Recipe	κ	Group	Sparsity (%)	View tail (%)	Grid tail (%)
A0	—	FFN	< 0.01	0.1113	0.0000
A0	—	Attention	< 0.01	2.3929	0.4632
A4-OL1	0	FFN	67.22	0.0003	0.0000
A4-OL1	0	Attention	< 0.01	5.1717	0.6162
A4-OL1	0.05	FFN	86.71	0.0003	0.0000
A4-OL1	0.05	Attention	< 0.01	5.6601	0.7607
A4-OL1	0.5	FFN	99.31	0.0001	0.0000
A4-OL1	0.5	Attention	0.22	0.0003	0.0000
A7-OL1	0	FFN	64.52	0.0063	0.0000
A7-OL1	0	Attention	< 0.01	5.4657	0.4648
A7-OL1	0.05	FFN	75.29	0.0106	0.0000
A7-OL1	0.05	Attention	7.26	6.6719	0.6932
A7-OL1	0.5	FFN	93.64	0.0037	0.0000
A7-OL1	0.5	Attention	95.60	0.2396	0.0009

Table 10: Entirely zero projection-input rows and mean nonzero coordinates per row (NNZ), separately at h and z , for the 14M $\kappa = 0.5$ endpoints. Layer indices start at zero. All entries use actual BF16 operands in the untimed K050 diagnostic; they are not the pooled FP16 h, m histograms.

Recipe	Layer	$h = 0$ rows (%)	Mean NNZ h	$z = 0$ rows (%)	Mean NNZ z
A4-OL1	0	12.45	1.729	90.34	0.116
A4-OL1	1	99.63	0.004	100.00	0.000
A4-OL1	2	100.00	0.000	100.00	0.000
A4-OL1	3	99.91	0.001	100.00	0.000
A4-OL1	4	99.59	0.011	92.59	0.074
A4-OL1	5	90.77	0.114	100.00	0.000
A7-OL1	0	22.60	2.022	69.09	0.405
A7-OL1	1	85.01	0.182	96.38	0.045
A7-OL1	2	75.10	0.296	95.72	0.047
A7-OL1	3	78.99	0.263	87.77	0.123
A7-OL1	4	62.28	0.526	98.24	0.018
A7-OL1	5	54.77	0.536	100.00	0.000

Exact-zero mass and nonzero activation distributions

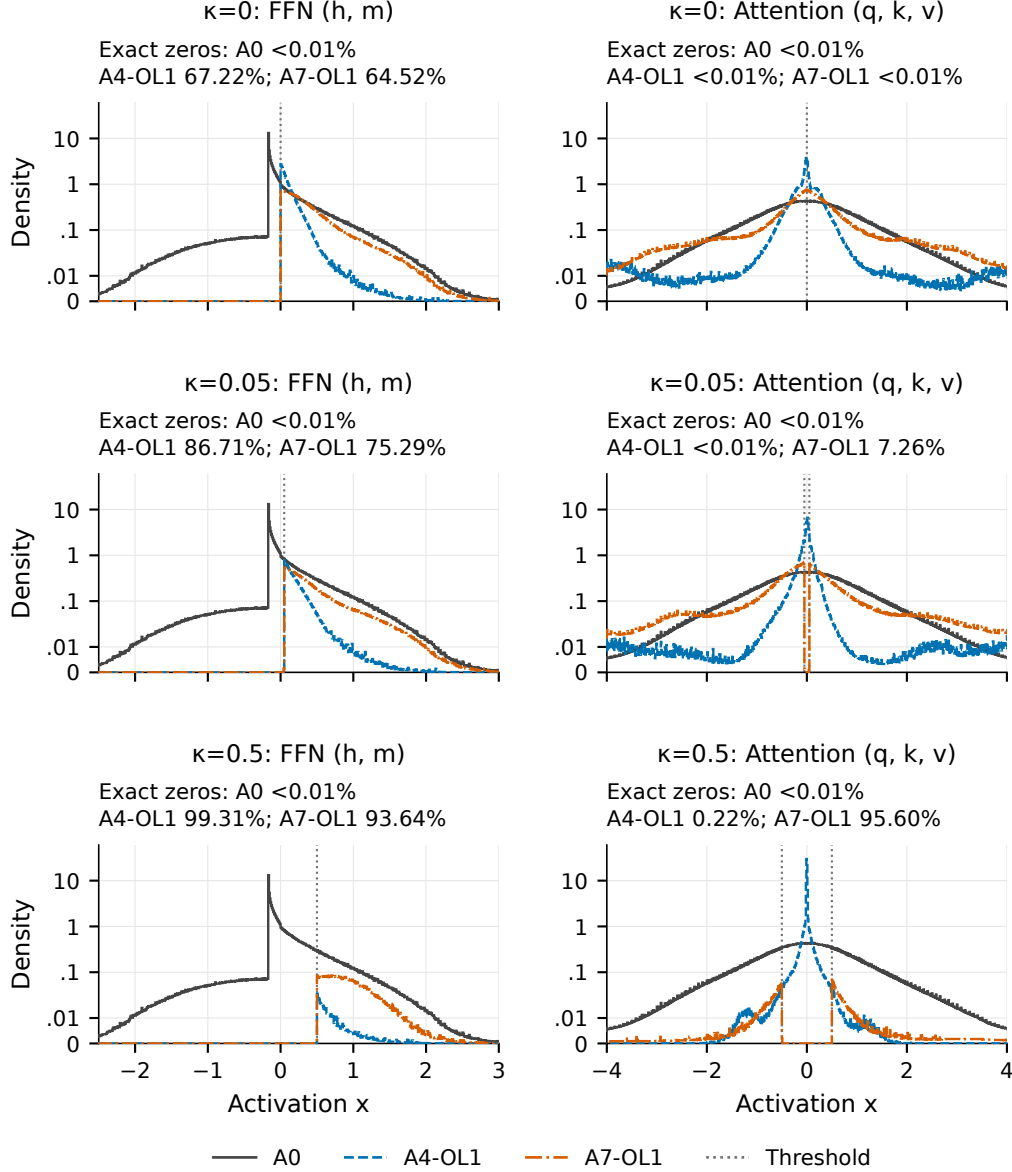


Figure 7: **Interventions reshape FFN and attention distributions differently. Exact zeros are reported above each panel and excluded from the nonzero curves.** Columns pool FFN (h, m) and attention (q, k, v) activations; rows use trained thresholds $\kappa = 0, 0.05, 0.5$. A0 is repeated across rows. Counts are pooled across all layers and validation blocks, weighting $h : m$ by 80:20 and $q : k : v$ equally. Density divides nonzero bin counts by all captured elements and bin width, without renormalizing the visible range. The vertical scale is symlog, linear below 0.01 and logarithmic above it; plotted areas do not represent probability mass. Dotted lines mark $+\kappa$ for A4/A7 FFN thresholds and $\pm\kappa$ for A7 attention thresholds. Q/K are measured after RoPE. A4 leaves q, k, v without direct thresholding or pressure; A7 targets them directly. The seven checkpoints use no additional post-hoc clipping.

Quality-sparsity trade-offs across Pythia sizes

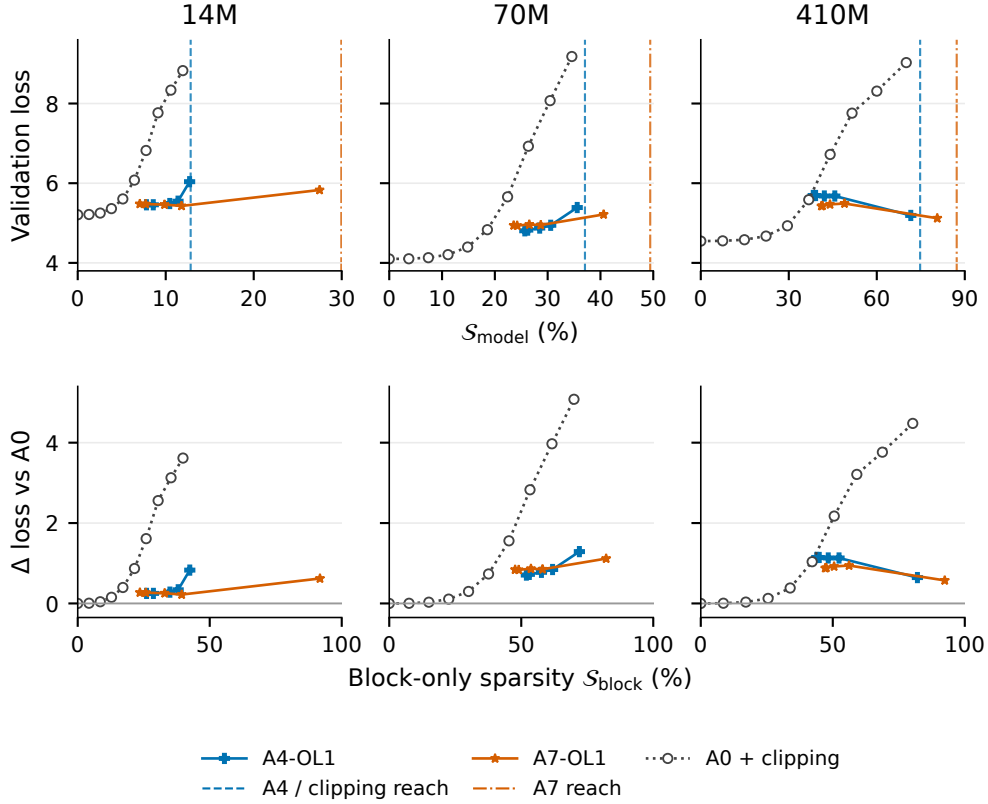


Figure 8: **The high-threshold A7-OL1 advantage persists across model sizes.** Columns compare 14M, 70M and 410M. Solid curves connect separately trained A4-OL1 and A7-OL1 checkpoints at $\kappa = 0, 0.01, 0.05, 0.1, 0.5$; dotted open-marker curves show A0 post-hoc clipping at a, m, h, z for targets $p = 0, 0.1, \dots, 0.9$. The 60 evaluations appear in both rows. Top: validation loss versus $100S_{\text{model}}$; vertical guides indicate A4/clipping-site and A7 architectural reach. Bottom: loss minus unmodified same-size A0 versus $100S_{\text{block}}$, where $S_{\text{block}} = S_{\text{model}}/R_{\text{arch}}(\text{A7})$ for **every curve**. The A7 reference reach values are 29.95%, 49.42% and 87.25%; the A4 guide is not the normalization denominator. At $\kappa = 0.5$, A7-OL1 improves both axes over A4-OL1 at all sizes; the ordering at smaller thresholds varies. Table 3 gives the boundary-threshold comparisons. These are complete-recipe comparisons with matched token budgets, unequal tokens per parameter and a lower 410M learning rate.

D.3 COMPLETE CROSS-SIZE CURVES AND NORMALIZATION

Figure 8 retains all selected A4-OL1/A7-OL1 thresholds and A0 clipping evaluations at every size, including unfavorable settings. The common A7 reach denominator equals the block-only normalization for this graph; it removes dense-head share, not changes in block-operation weights. The higher 410M A0 loss than 70M’s under the shared token budget is retained.

D.4 OPERATION CONTRIBUTIONS TO MODEL-WIDE SPARSITY

An operation’s contribution to S_{model} is its zero-product count divided by the full model product count. Summing these contributions recovers the model-wide fraction; summing local activation

percentages would not. Table 11 gives this decomposition at $\kappa = 0.5$. At 14M, extending from A4-OL1 to A7-OL1 adds attention opportunity while reducing the contribution from some projections. At larger sizes, projection work carries more weight in the workload, so an equal change in activation sparsity need not yield an equal change in model-wide sparsity. Figure 9 displays the operation-level view at the same threshold and all three sizes.

Table 11: Contributions to $100S_{\text{model}}$ in percentage points at $\kappa = 0.5$. All six operations share the same model denominator within each size. The dense LM head contributes to that denominator but receives no zero credit. Differences in rounded entries can affect the displayed sum.

Operation	14M		70M		410M	
	A4-OL1	A7-OL1	A4-OL1	A7-OL1	A4-OL1	A7-OL1
QKV	3.173	2.430	8.820	6.231	17.563	16.992
QK	0.000	8.324	0.000	5.052	0.000	5.206
PV	0.057	8.449	0.011	5.371	0.008	5.858
Attention output	1.069	1.069	3.087	3.084	6.186	6.187
FFN up	4.139	2.939	11.329	8.526	22.938	21.503
FFN down	4.275	4.272	12.350	12.338	24.895	24.869
Total	12.713	27.483	35.596	40.602	71.591	80.616

Operation contributions at $\kappa = .5$

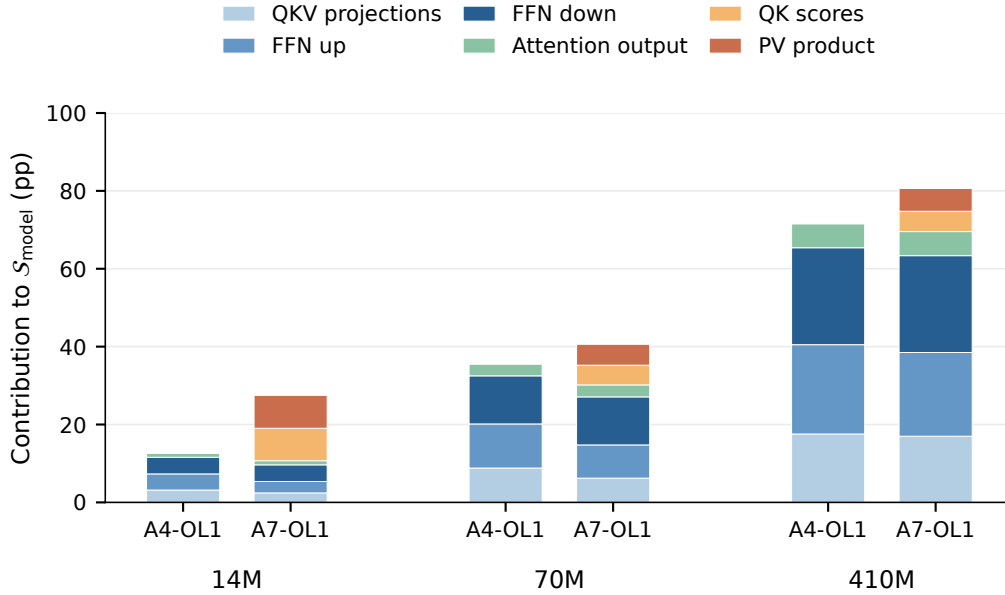


Figure 9: Operation accounting for A4-OL1 and A7-OL1 at $\kappa = 0.5$ across 14M, 70M and 410M. Zero-operand products are counted over causally valid positions and pooled across all 338 validation blocks. Each contribution uses the full model denominator, including the dense LM head without zero credit. At 14M, attention contributes more under A7-OL1, outweighing smaller contributions from several projections. The bars show logical multiplication opportunities, not runtime savings.

D.5 COMPLETE POST-HOC CLIPPING TRAJECTORIES

The complete release contains 540 evaluations: ten clipping targets for each of the 54 trained checkpoints. It joins 350 new evaluations with 190 retained evaluations matched by checkpoint content hash. Calibration uses the first ten complete training blocks in source order, separately for each

layer’s a, m, h, z activations. Evaluation zeros values satisfying $|x| \leq t$ after the trained nonlinearities, without changing weights or A7’s internal attention nonlinearities. The target p controls a calibration quantile; it is not the achieved model-wide sparsity. In particular, an existing zero mass can make several targets identical.

Figures 10–12 retain the full loss range and every target, including dominated points and ties. Each path begins at its actually measured $p = 0$ evaluation. Small numerical differences from the canonical training endpoint can change strict nondominance without implying a substantive improvement. Across all checkpoints, the absolute $p = 0$ discrepancy is at most 1.81×10^{-4} in loss and 0.000908 sparsity percentage points. Checking dominance beyond the observed same-size drift on both axes still finds a trained endpoint with lower loss and higher sparsity than each A0 clipping evaluation at $p \geq 0.5$, for every size. This is a numerical sensitivity check, not a confidence interval or an estimate of training variability. Numerical frontier memberships are stored separately from the lines connecting evaluated settings. All evaluations use complete validation with FP32 parameters, FP16 autocast, eager attention, batch one and $T = 2,048$.

14M training and post-hoc clipping

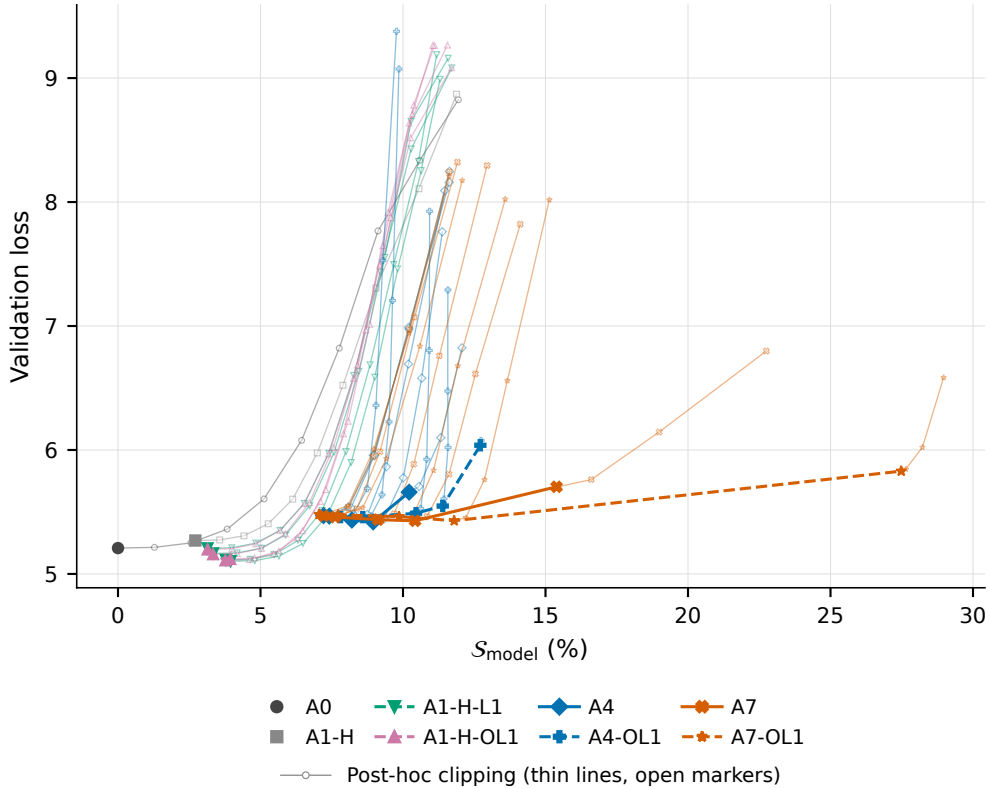


Figure 10: Complete 14M training and post-hoc clipping results: 30 trained endpoints and 300 clipping evaluations. Large filled markers denote trained endpoints; bold lines connect their settings, solid for unpressured recipes and dashed for L1/OL1. Each faint open-marker path follows one fixed checkpoint through $p = 0, 0.1, \dots, 0.9$. Colors and symbols identify its source recipe. The full loss range is shown. Connections order evaluated settings and do not interpolate attainable models or fitted Pareto envelopes.

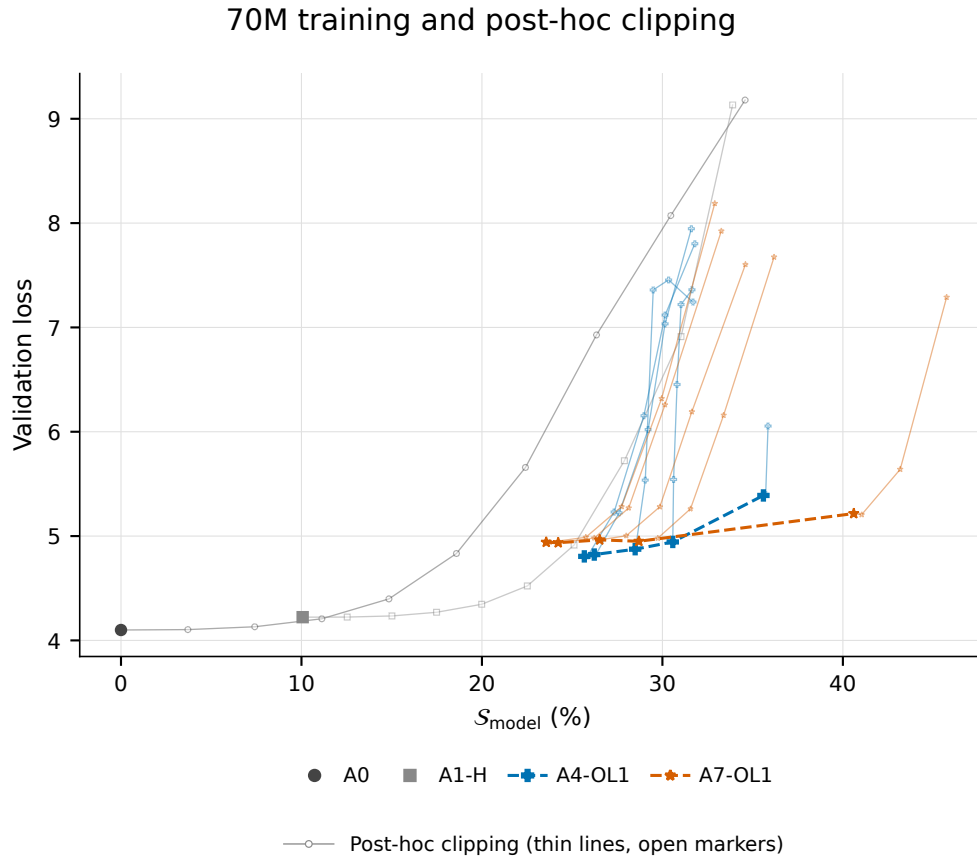


Figure 11: Complete 70M training and post-hoc clipping results: 12 trained endpoints and 120 clipping evaluations. Large filled markers and bold lines identify the trained recipes; faint open-marker paths follow each checkpoint through ten increasing clipping targets. All points, including overlaps and high-loss settings, are retained. The evaluation and line conventions match Figure 10.

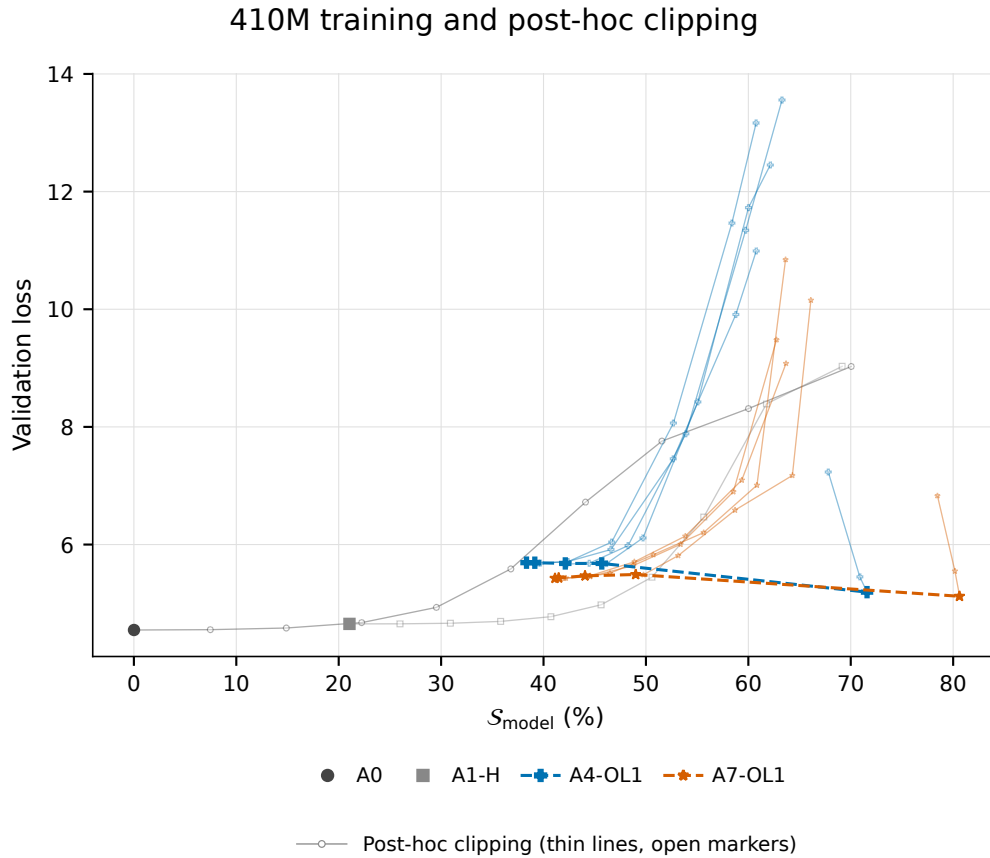


Figure 12: Complete 410M training and post-hoc clipping results: 12 trained endpoints and 120 clipping evaluations. Large filled markers and bold lines identify the trained recipes; faint open-marker paths follow each checkpoint through ten increasing clipping targets. All points, including overlaps and high-loss settings, are retained. The evaluation and line conventions match Figure 10.

D.6 KERNEL QUALIFICATION AND ABLATIONS

The unchanged Sakana/TwELL control achieves $1.3001\times$ on its supported SparseLM-0.5B/H100 workload. Its raw projection primitive at the Pythia-14M width $N = 128$ instead gives relative L2 error 0.69055, exceeding the 0.02 bound; that test stops before full-model Pythia timing. This is a compatibility test, not an unchanged upstream speedup comparison on our model. The later P0 adapter changes signed-value packing, shape tails and numerical handling for Pythia. It retains dense attention and uses native projection fallback on A0. Table 12 reports this adapted baseline.

Historical proposals are replayed on four representative checkpoints; the plotted incumbent uses fixed A7-OL1, $\kappa = 0.5$. Final implementation and ablation summaries retain the 30-checkpoint manuscript cohort from the historical 35-checkpoint sweep. Five A4 checkpoints with pressure only at h remain outside the main cohort because they implement a distinct objective. Their original configurations, inherited capture code, initialization, schedule and checkpoint identities were reconciled against the corrected four-site runs. Across the five matched thresholds, four-site pressure raises diagnostic loss by 0.2425–0.3196 relative to h -only pressure. Its sparsity change is -0.599 pp at zero threshold, $+0.001$ pp at 0.01, and $+1.421$ – $+2.486$ pp at larger thresholds. Adding pressure sites changes each old site–layer coefficient from $1/6$ to $1/24$, so this historical comparison does not hold per-site pressure fixed. Search progress and the final sweep are separate measurements, explaining the small difference between their $1.7830\times$ and $1.7832\times$ maxima.

Matched retrospective search and native-relative association

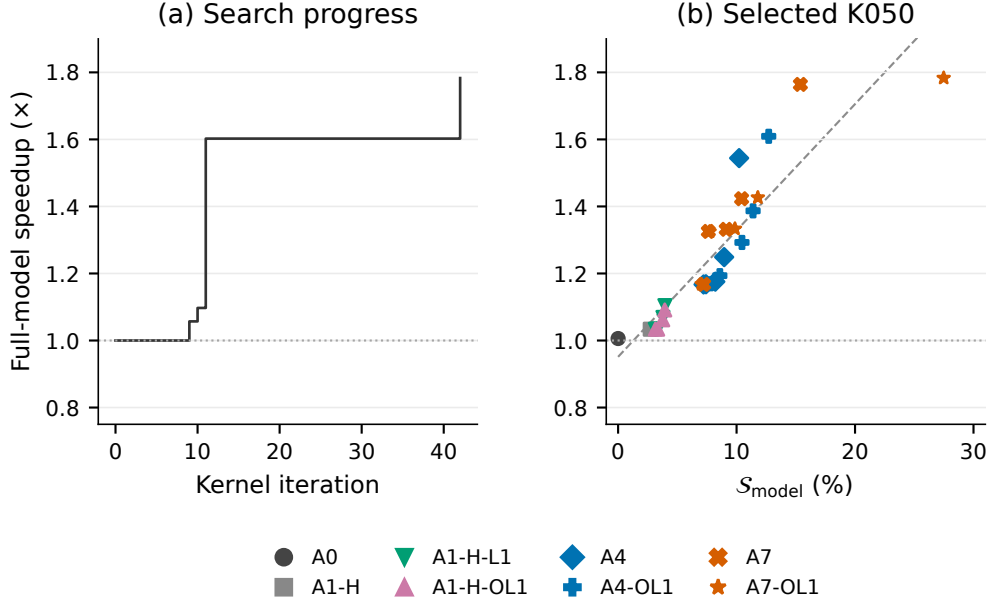


Figure 13: Retained development history and descriptive native-relative association. (a) Best qualified incumbent on the fixed A7-OL1, $\kappa = 0.5$ search checkpoint: $1.6024\times$ by iteration 11 and $1.7830\times$ at iteration 42. The later attention-dense ablation is not substituted into this history. (b) Selected K050 on all 30 canonical checkpoints, with an OLS fit and intercept ($R^2 = 0.8167$). Sparsity uses FP16 counts and timing uses BF16; the axes have restricted ranges. Both panels use the matched RTX5090, batch-one, full-sequence native reference specified in the text.

The coding-agent configuration records *gpt-6-astra* as the default model, not the model served for every request. This is one human-guided development trajectory and does not compare agent-assisted with human-only kernel development. Kernel measurements use Python 3.12.3, PyTorch 2.11.0+cu128, Transformers 5.12.1, CUDA 12.8 and nvcc 12.8.93. The supplementary protocol records dependency and frozen-source hashes.

Every candidate must produce finite outputs, with each logit satisfying $|y - \hat{y}| \leq 0.25 + 0.02|\hat{y}|$, per-block relative output L2 error at most 0.02, and absolute pooled loss difference at most 0.001. Here $\hat{y}$ is the unmodified native eager reference. All three fresh processes must pass on all 338 validation blocks. Qualification is bounded numerical agreement on these inputs, not bitwise equivalence. Failed candidates are excluded from qualified speedup summaries. P0, an early projection adapter derived from the released kernels of Cetin et al. (2026), qualifies on only four checkpoints. Its average is not comparable to a 30-checkpoint mean or an unchanged upstream benchmark.

Timing uses the same checkpoint, inputs and CUDA-graph scaffold for native PyTorch/SDPA and each candidate. For each checkpoint, we take the geometric mean of 1,344 paired host-latency ratios (native divided by candidate): 64 inputs, seven randomized paired passes, and three fresh processes. Compilation, static weight preparation, graph capture and equal input staging are excluded. Preprocessing that depends on activations, embeddings, every block, final normalization and full vocabulary logits are included. There is no activation cache. Cohort summaries give each checkpoint equal weight.

K050 combines normalization and RoPE fusion, zero-fragment tensor-core bypass and sparse-row projection paths, using CUDA, CUTLASS, FlashAttention and PyTorch components. K049 is its searched predecessor; K050 adds fusion of the two input LayerNorms while preserving their separate affine parameters and thresholds. The all-skips-off control retains fusion while disabling both attention and projection skips; the attention-dense control retains projection skips. Their ratios are formed from separately measured, native-normalized timings. K050’s sparse paths improve 14 of 30 checkpoints relative to all-skips-off, with geometric mean ratio $1.0432\times$. Disabling attention skipping improves all 30: at the search checkpoint, instrumentation confirms that skipping bypasses 58.0% of eligible QK and 67.2% of PV tensor-core matrix multiply-accumulate (MMA) instructions, yet its overhead exceeds the saved time. Scalar zero-product fractions and skipped MMA fractions are different estimands.

The attention path tests whether either warp-distributed operand fragment contains only zeros, and issues an MMA only when both fragments contain a nonzero. Partial sparsity can therefore leave an entire instruction active. Zero detection, warp votes and branches recur inside the kernel; causal masking and softmax still execute. A zero QK score does not imply a zero softmax probability. The measurements do not identify a profitable joint-zero pattern or break-even threshold. They show that the skipped work does not repay these costs for the measured 14M shapes and hardware.

Table 12: Numerically qualified full-model speedups against native SDPA CUDA graphs. GM is the equal-checkpoint geometric mean; the range is across qualified checkpoints, not a confidence interval. All conditions use one RTX5090, BF16, batch one, 2,048 tokens and full vocabulary logits. P0’s four-checkpoint subset is shown only to document its limited qualification.

Implementation	Qualified	GM speedup	Range
K049	30/30	1.1141	0.9550–1.5284
K050	30/30	1.2340	1.0059–1.7832
K050, attention dense	30/30	1.2506	1.0185–1.8022
K050, all skips off	30/30	1.1829	1.0503–1.3628
P0 (qualified subset only)	4/30	0.8910	0.8180–1.0012

D.6.1 ABSOLUTE LATENCIES AND ASSOCIATION CHECKS

Table 13 retains actual host latencies: for each implementation and checkpoint, the geometric mean of the three process medians, each over 448 timed sequences. The native column uses K050’s paired native measurements; the ablation columns have their own native-paired processes. A ratio of these median summaries need not equal the reported geometric mean of paired ratios. The last column instead divides K050’s native-relative speedup by all-skips-off’s native-relative speedup. BF16 validation losses accompany these timings; canonical training tables retain their original FP16 loss/count pass.

The association checks in Table 14 fit an unweighted checkpoint-level line with an intercept. Removing the searched checkpoint or a recipe family tests descriptive sensitivity; it does not create a held-out prediction study. Within-family fits use only four or five points, and their larger R^2 values

Table 13: Matched 14M host latency summaries in milliseconds, BF16 validation loss and incremental sparse-path factor. Dose is λ for local pressure and κ for A4/A7. All four implementations qualify on every row. These are unmodified trained checkpoints; clipped endpoints have no corresponding latency measurements. The sparse factor is defined in the text, not computed by dividing the displayed rounded latency columns.

ID	Recipe	Dose	BF16 loss	Native	Skips off	K050	Attn. dense	Factor
c01	A0	—	5.2100	0.6555	0.6213	0.6516	0.6437	0.9535
c02	A1-H	—	5.2712	0.6539	0.6224	0.6328	0.6242	0.9835
c03	A1-H-L1	0.05	5.2082	0.6541	0.6221	0.6329	0.6237	0.9828
c04	A1-H-L1	0.1	5.1668	0.6539	0.6226	0.6324	0.6240	0.9843
c05	A1-H-L1	0.5	5.1136	0.6541	0.6220	0.6120	0.6044	1.0162
c06	A1-H-L1	1	5.1040	0.6543	0.6231	0.5922	0.5838	1.0512
c07	A1-H-OL1	0.05	5.1991	0.6540	0.6225	0.6326	0.6254	0.9843
c08	A1-H-OL1	0.1	5.1609	0.6540	0.6223	0.6331	0.6247	0.9833
c09	A1-H-OL1	0.5	5.1114	0.6542	0.6230	0.6159	0.6082	1.0106
c10	A1-H-OL1	1	5.1228	0.6540	0.6219	0.5989	0.5909	1.0372
c11	A4	0	5.4727	0.7399	0.6234	0.6338	0.6250	0.9820
c12	A4	0.01	5.4694	0.7402	0.6234	0.6339	0.6251	0.9839
c13	A4	0.05	5.4364	0.7400	0.6233	0.6295	0.6213	0.9898
c14	A4	0.1	5.4218	0.7399	0.6230	0.5917	0.5836	1.0511
c15	A4	0.5	5.6623	0.7394	0.6242	0.4790	0.4713	1.3022
c16	A4-OL1	0	5.4598	0.7402	0.6229	0.6331	0.6247	0.9842
c17	A4-OL1	0.01	5.4602	0.7401	0.6239	0.6191	0.6108	1.0054
c18	A4-OL1	0.05	5.4922	0.7401	0.6236	0.5706	0.5628	1.0885
c19	A4-OL1	0.1	5.5504	0.7399	0.6233	0.5325	0.5249	1.1674
c20	A4-OL1	0.5	6.0403	0.7394	0.6227	0.4591	0.4511	1.3542
c21	A7	0	5.4704	0.7400	0.6227	0.6339	0.6260	0.9824
c22	A7	0.01	5.4611	0.8460	0.6212	0.6382	0.6306	0.9734
c23	A7	0.05	5.4405	0.8458	0.6208	0.6352	0.6267	0.9773
c24	A7	0.1	5.4309	0.8453	0.6209	0.5934	0.5858	1.0443
c25	A7	0.5	5.7076	0.8449	0.6211	0.4789	0.4714	1.2955
c26	A7-OL1	0	5.4824	0.7400	0.6231	0.6337	0.6255	0.9831
c27	A7-OL1	0.01	5.4778	0.8456	0.6213	0.6384	0.6308	0.9725
c28	A7-OL1	0.05	5.4651	0.8459	0.6214	0.6346	0.6274	0.9782
c29	A7-OL1	0.1	5.4315	0.8459	0.6210	0.5920	0.5847	1.0473
c30	A7-OL1	0.5	5.8313	0.8441	0.6210	0.4731	0.4679	1.3112

do not remove the differences among recipes. All fits compare canonical FP16 $\mathcal{S}_{\text{model}}$ with BF16 timing. The retained BF16 scalar counter excludes probability-underflow zeros and is only a lower bound, so it cannot replace the canonical counter to resolve that precision mismatch.

Table 14: Descriptive association sensitivity. Each column regresses its timing ratio on canonical FP16 model-wide sparsity with an intercept. Native relative uses K050/native; sparse factor uses the incremental definition above. Subsets are retrospective and overlapping, with no independent training seeds.

Checkpoint subset	n	Native-relative R^2	Sparse-factor R^2
All checkpoints	30	0.8167	0.5037
Within A1-H-L1	4	0.8944	0.8908
Within A1-H-OL1	4	0.8888	0.8921
Within A4	5	0.8353	0.8346
Within A4-OL1	5	0.9087	0.9095
Within A7	5	0.9528	0.9356
Within A7-OL1	5	0.9403	0.9825
Without A0	29	0.8104	0.4943
Without A1-H	29	0.8101	0.4989
Without A1-H-L1	26	0.7937	0.5077
Without A1-H-OL1	26	0.7929	0.5007
Without A4	25	0.8497	0.5685
Without A4-OL1	25	0.8319	0.5322
Without A7	25	0.8401	0.4977
Without A7-OL1	25	0.8520	0.4910
Without search checkpoint	29	0.8521	0.4080

D.7 EVIDENCE AND DATA ACCOMPANYING THE DRAFT

The supplementary directory `supplementary-data/` contains all 54 training records with integer operation counts, all 540 clipping evaluations and frontier memberships, seven signed histograms with zero and tail counts, figure coordinates, and the 30-checkpoint runtime reduction. The included README defines fields and units. Copy hashes are recorded in `SOURCES.json`; `protocol.json` gives data preparation, training settings and implementation identities. Paths refer to the accompanying repository; the supplementary directory is not a standalone checkpoint or kernel release. The `revision-audit/` subdirectory adds the retrospective quality-budget reduction, absolute runtime summaries and sensitivity checks, per-layer projection-input row statistics, and historical-pressure identity audit. Raw identifiers remain unchanged: $\mathcal{S}_{\text{model}}$ corresponds to `R_model`, $\mathcal{S}_{\text{block}}$ to `R_block`, and architectural reach R_{arch} to the existing `R_model_max` fields.

The accompanying repository includes the figure and table scripts. The Analysis 019 README gives commands for regenerating the revised visuals from the retained numerical records, separate from commands that require the original raw run artifacts. The draft README gives the LaTeX build commands. Checkpoint weights and runtime compilation dependencies are not bundled in the supplementary measurement directory; these commands do not claim to reproduce training or GPU measurements from that directory alone.